

LIN7076 – Foundations of Computational Linguistics

Formal languages

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Deep intuitions about well-formedness

- (1) a. Colorless green ideas sleep furiously.
- b. * Furiously colorless green sleep ideas.

'Twas brillig, and the slithy toves
Did gyre and gimble in the wabe:
All mimsy were the borogoves,
And the mome raths outgrabe.
“Beware the Jabberwock, my son!
The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
The frumious Bandersnatch!”

Jabberwocky, Lewis Carroll

Can we come up with simple rules summarizing a language or a fragment thereof?

- We'll take a break from statistics and algebra to talk a bit about formalizing language structure.
- We'll explore two main approaches: **regular expressions**, **context-free grammars**.
- We'll discuss how these can be used for concrete applications: find-and-replace, parsing and structure-mining.
- We'll also review some aspects of natural language that are fundamentally challenging for these approaches.

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Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(2) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(3) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(4) a. * Jo eats often apples. b. Jo often eats apples.

- All these generalizations are independent of meaning—they are **syntactic**, and sometimes language-dependent.
- Violating them causes ungrammaticality/**ill-formedness**.

Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(5) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(6) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

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- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(8) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(9) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(10) a. * Jo eats often apples. b. Jo often eats apples.

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Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(11) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(12) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(13) a. * Jo eats often apples. b. Jo often eats apples.

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(14) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(15) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

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Examples of infelicity

- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(17) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(18) a. # A sun is shining. b. The sun is shining.
 ↪ There's at least one sun. ↪ There's a unique sun.

- But should not take new information for granted:

(19) a. ?? Jo will need to bring her alligator to the vet. b. Jo has an alligator, and will need to bring it to the vet.

- These generalizations depend on meaning (semantic) and largely language-independent. Violating them causes infelicity.

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- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(20) a. # Jo is inside and outside. b. # Jo is inside or outside.

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(21) a. # A sun is shining. b. The sun is shining.
 $\rightsquigarrow$ There's at least one sun. $\rightsquigarrow$ There's a unique sun.

- But should not take new information for granted:

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(23) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(24) a. # A sun is shining. b. The sun is shining.
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- But should not take new information for granted:

(25) a. ?? Jo will need to bring **her**
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- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(26) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(27) a. # A sun is shining. b. The sun is shining.
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(28) a. ?? Jo will need to bring **her**
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(30) a. # A sun is shining. b. The sun is shining.
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- But should not take new information for granted:

(31) a. ?? Jo will need to bring **her**
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(33) a. # A sun is shining.
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Formalizing well-formedness

- Today we'll **focus on modeling grammaticality/well-formedness**.
- A language can be modeled as the set of all the sentences that are well-formed in that language—independently of meaning.
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Other possible specifications of a “language”

- More abstractly, a language can be seen as the set of well-formed sequences of syntactic categories (instead of words instantiating these categories).
- In that case, English = {Det Noun Verb, Adj Adj Noun Verb Adv, ...}. How big is this set?
- More narrowly, a language can be seen as the set of well-formed sequences of phonological elements (e.g. consonants C, and vowels V¹)
- English is hard to illustrate, so let's take Japanese whose building blocks are mostly CV syllables plus occasional nasals. In that case Japanese = {CV, CVCV, CVCVCV, CVnCV, CVCVn, ...}.
- In all these cases, we are interested in **well-formed units of a language** (strings of words; strings of syntactic labels; strings of phonemes...).

¹One can choose more specific categories too!

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The need for recursivity

- To model natural languages, formal languages need to generate **an infinite number of outputs from finite means**.
- Finite means:
 - A finite alphabet Σ of primitive “words” (lexical items, or syntactic categories).
 - A finite set of rules with some degree of recursivity.
- Recursivity is the **capacity for a rule to refer to itself**. Because recursive rules may apply an arbitrary number of times, an infinity of outputs can be generated.
- Concrete example: a folder-based file system is inherently recursive. Any **folder** may contain only **files** (“base case”), or, at least one other **folder** (“recursive case”).
- More generally, anything that displays an **arborescent structure** will be recursive!

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A toy example: intensified adjectival phrases

- Suppose our alphabet Σ is { a, the, very, tall, girl, boy }. What kind of well-formed phrases can we build out of this alphabet?
- *the boy* and *the girl* are well-formed.
- *a boy* and *a girl* are well-formed.
- *the/a tall boy/girl* are well-formed.
- *the/a very tall boy/girl* are well-formed.
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Regular expressions

- Let Σ be an alphabet. We call the members of Σ **words** (they can be “real” words like *cat*, syntactic labels like *Det*, phonological units like *V...*).
- Regular expressions (**regexes**) are used to specify **word patterns**.
- Regexes are built out of consist of **constants**, which denote sets of words, and **operator symbols**, which denote operations over these sets.
- Constants: $\emptyset$ (the empty regex), and $\{\epsilon\}$ (the empty-word regex) are regexes. For any w in Σ , $\{w\}$ is a regex.
- Operations: given R and S two regexes...
 - RS is the pairwise **concatenation** of R and S and is a regex.
 - $R|S$ is the **union** of R and S and is a regex.
 - R^* is the set containing ϵ and all possible concatenations of elements of R (“**Kleene star**”) and is a regex.
- We’ll use parentheses to signal the order of precedence between these operations.

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- **Concatenating** all these parts together form the desired pattern: (*a|the*)(*very**)*tall| ϵ*)(*boy|girl*).
- What is the issue with the regex (*a|the*)(*very**)(*tall| ϵ*)(*boy|girl*)?

Useful shorthands

- One can define useful **shorthands** from the three core operations (concatenation, union, Kleene star). If R and S are regexes:
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- Mining a dataset for specific verbs and all their **possible inflections** (including noisy/erroneous ones!), for instance **CHILDES** when studying the acquisition of verbal predicates.
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- Automata have **states**, represented with **circles**. There's one **initial** state (where you start), and at least one **accepting** state (where you want to end, signaled with a double circle).
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- More generally, regex **concatenation** is represented by **arrows between states**.

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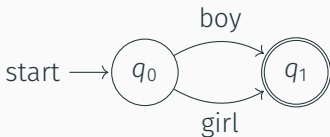
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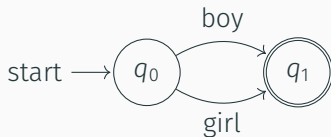
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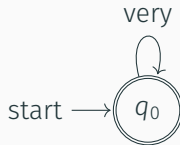
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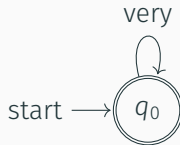
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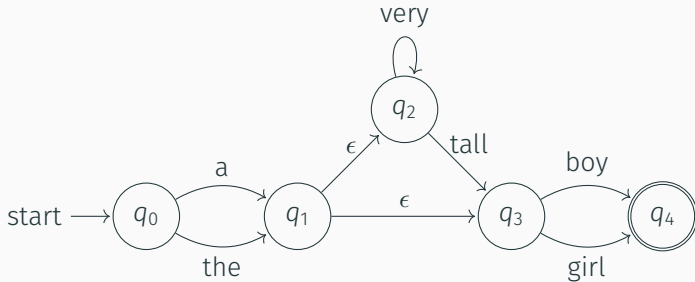
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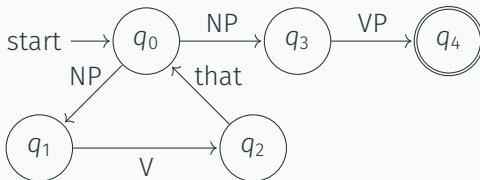
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- It meant that first we want NP V that (an embedding sentence) zero or more times, and then we finish by NP VP (the last embedded sentence).
- What's the automaton for this?



Linguistic challenges for regular expressions and automata

Structural ambiguities

- Compare:

- (35) a. Tall boys and girls. c. Tall boys and short girls.
 b. Tall boys and tall girls. d. Boys and tall girls.

- When an adjective precedes conjoined NPs, as in (35a), it's **ambiguous whether the adjective applies to both NPs** (as in (35b)), **or to the first NP only** (with a meaning compatible with (35c)).
- This ambiguity disappears if the adjective follows the first and precedes the second NP, as in (35d).
- Similar-ish ambiguities arise in sentences like (36) and (37).

(36) The photographer saw the bird on the hill.

(37) They are building blocks.

Structural ambiguities

- Compare:

- (38) a. Tall boys and girls. c. Tall boys and short girls.
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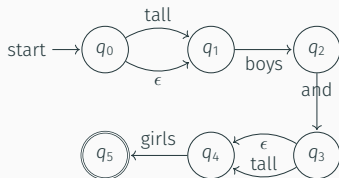
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Regexes/automata have a “weak” generative capacity

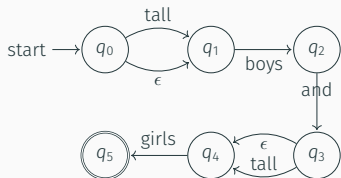
- Regexes/automata *can* match strings like *tall boys and girls*, *tall boys and tall girls* etc: $(\text{tall}|\epsilon)$ boys and $(\text{tall}|\epsilon)$ girls.



- But nothing in the traversal of the automata, or in the denotation of the regex (a sets of strings!) tells us that *tall boys and girls* can be understood either as *[tall [boys]] and [girls]* or as *[tall [boys and girls]]*!
- These devices capture the well-formedness of the string, but **nothing about its internal structure** (which influences meaning).
- In that sense, regexes and automata have a **weak generative capacity** w.r.t. structural ambiguities.

Regexes/automata have a “weak” generative capacity

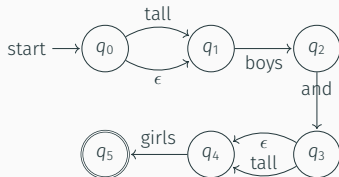
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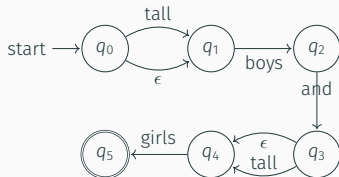
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Deeper challenge: center-embedding

- Let's consider nested subject-modifying, object-gap relative clauses:

- (47)
- a. The mouse is scared.
 - b. The mouse [that the cat chases] is scared.
 - c. The mouse [that the cat [that the dog barks at] chases] is scared.
 - d. The mouse [that the cat [that the dog [that the postman avoids] barks at] chases] is scared.
- The corresponding structures are: NP VP; NP that NP V VP; NP that NP that NP V V VP; NP that NP that NP that NP V V V VP... etc.
 - Can we synthesize this using one single regex?
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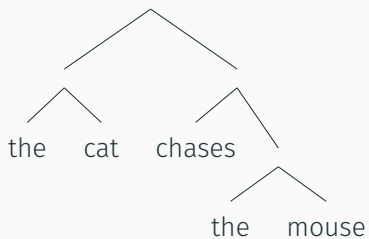
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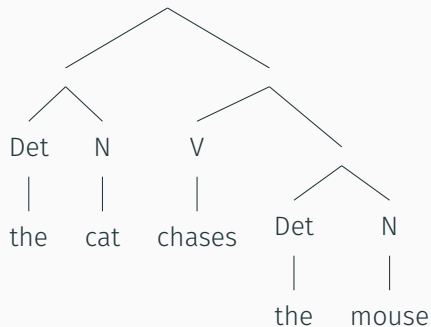
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Context-free grammars

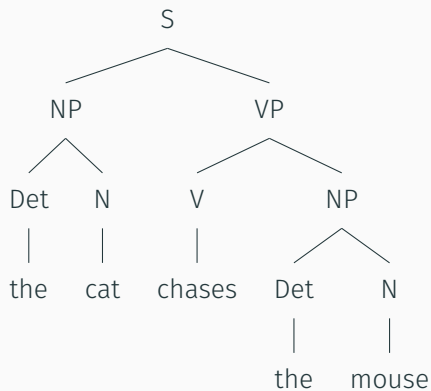
Syntax trees



Syntax trees



Syntax trees



Generating syntax trees

- One can use context-free grammar to generate syntax trees. A context-free grammar is defined as:
 - A finite set Σ of **terminals**, (leaves, i.e. actual words of the language).
 - A finite set V of **nonterminals** (intermediate nodes, i.e. phrase or category labels). V contains a special nonterminal S used to represent entire sentences (root).
 - A set R of **production rules**, i.e. input $\rightarrow$ output pairs, where inputs are single nonterminals (understood as mother nodes) and outputs are (non)terminals or concatenations thereof (understood as children node(s)). For instance, $\begin{array}{c} S \\ \swarrow \searrow \\ NP \quad VP \end{array}$ corresponds to the

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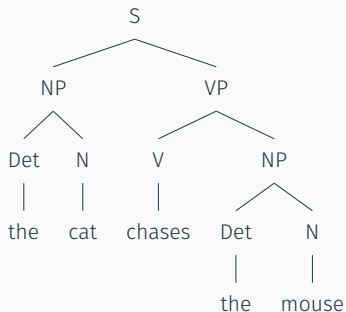
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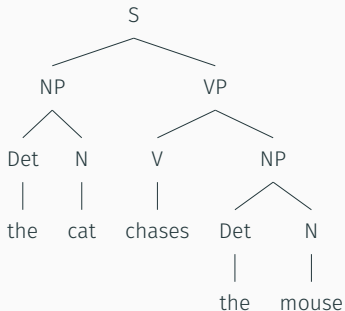
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Production rules practice



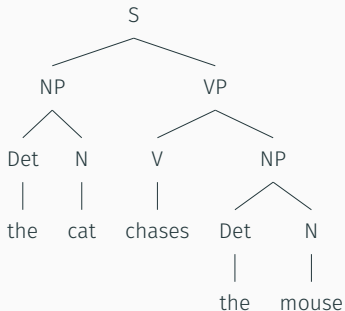
- What are the terminals?

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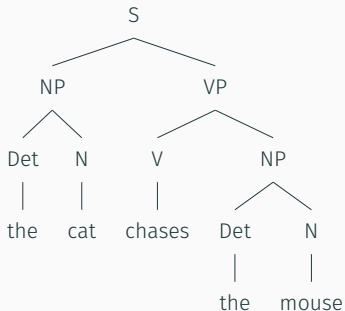
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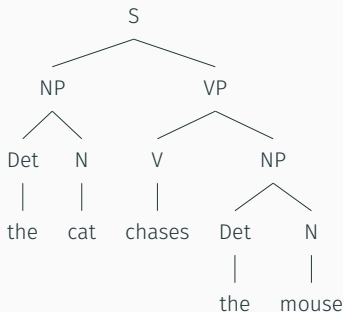
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- What else can we generate from those rules?

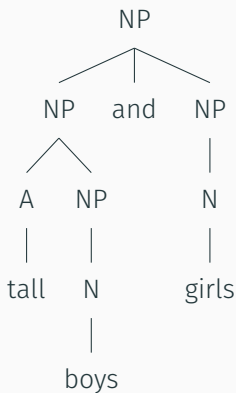
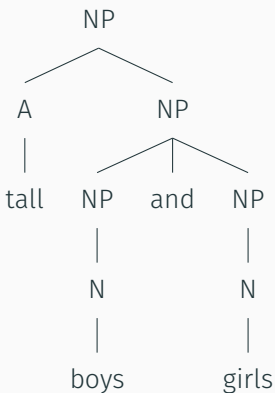
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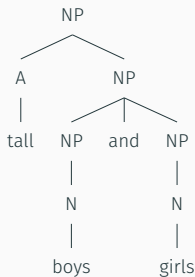
Structural ambiguity 1: adjective attachment

- Intuitively, which tree corresponds to which interpretation?



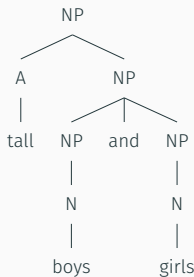
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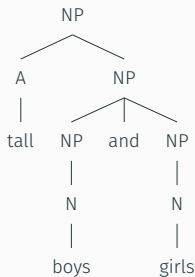
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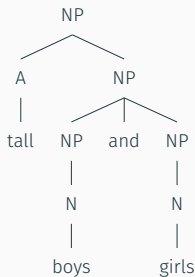
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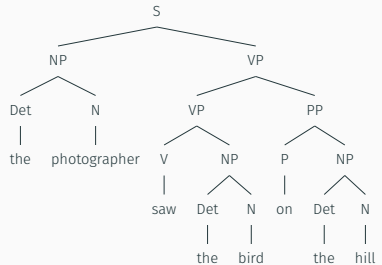
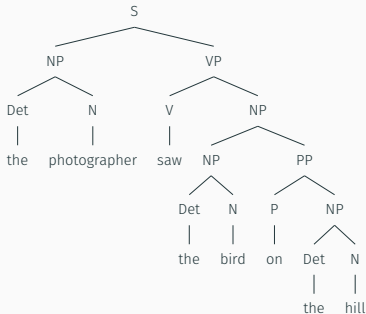
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Structural ambiguity 2: PP-attachment

(36) The photographer saw the bird on the hill.



- Which tree intuitively yields which reading?

Structural ambiguity 2: PP-attachment

- Consider the grammar below. Which reading(s) does it capture?

$S \rightarrow NP VP$

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$VP \rightarrow V NP$

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- The only rule in which PP is an output (daughter node) is $NP \rightarrow NP PP$. In that rule, PP's sister node is NP, which means PP attaches to an NP. It models the *bird on the hill*-reading.

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$Det \rightarrow the$

$N \rightarrow photographer, bird, hill$

$V \rightarrow saw$

$P \rightarrow on$

Structural ambiguity 2: PP-attachment

- Consider the grammar below. Which reading(s) does it capture?

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

$VP \rightarrow VP PP$

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- The only rule in which PP is an output (daughter node) is $VP \rightarrow VP PP$. In that rule, PP's sister node is VP, which means PP attaches to a VP. It models the *photographer on the hill*-reading.

Context-free grammar and structural ambiguities recap

- Context-free grammars primarily generate strings by **calling production rules** (i.e. branching), until all the outputs are terminals.
- But interestingly, keeping track of which production rules were used to generate the final string (and what was their exact timing), sometimes allows us to **derive distinct syntax trees for the same string**—cashing out semantic ambiguities.
- In that sense, context-free grammars have a **strong generative capacity** when it comes to adjective and PP attachment ambiguities.
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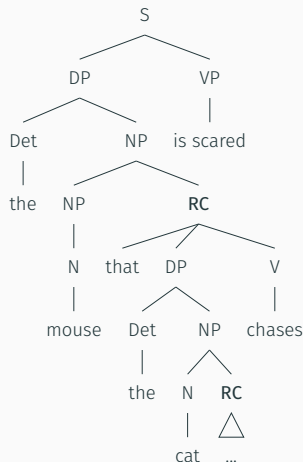
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Center-embedding: tree

- Recall the center-embedding pattern that regexes and automata could not capture: NP $\underbrace{(\text{that NP})^k (V)^k}_{\text{nested relative clauses}}$ VP, **for any** $k \geq 0$.

- We treat relative clauses (RCs) a bit like **complex adjectives**: they may attach to NPs, below the determiner.
- Trick: RCs themselves contain NPs, which themselves may contain other RCs! This makes RCs **recursive**: RCs may introduce new, nested RCs.
- Ultimately, this enables center-embedding.

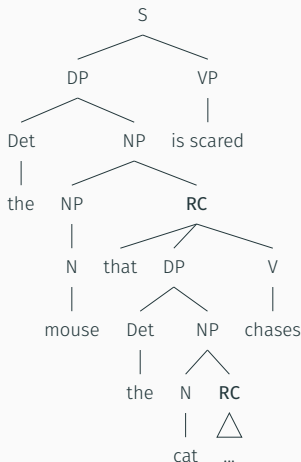


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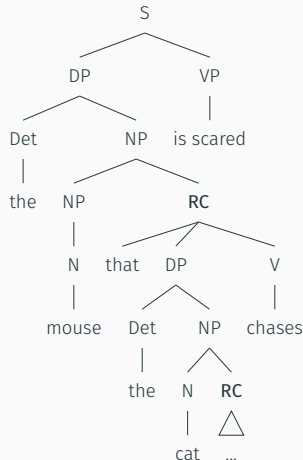
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Center-embedding: critical production rules

- Focusing on the nonterminal production rules:

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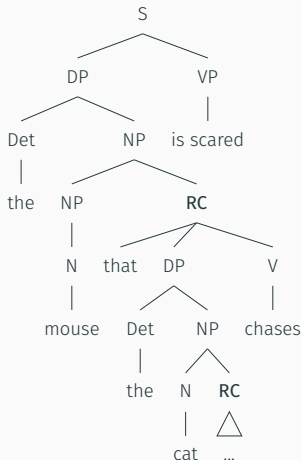
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$NP \rightarrow N$

$RC \rightarrow that DP V$

- NPs introduce RCs or simply yield a N (crucial base case, for the derivation to stop at some point!)
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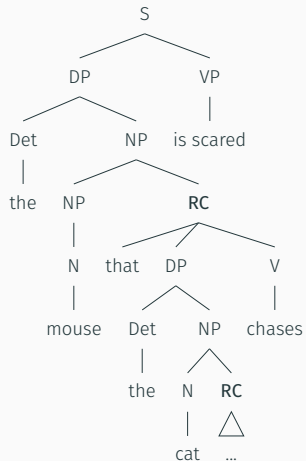
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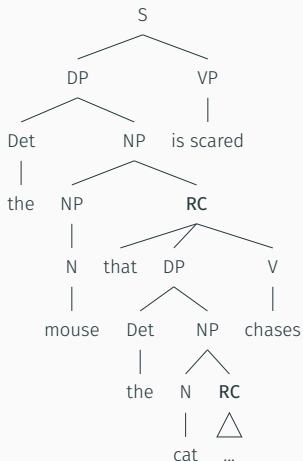
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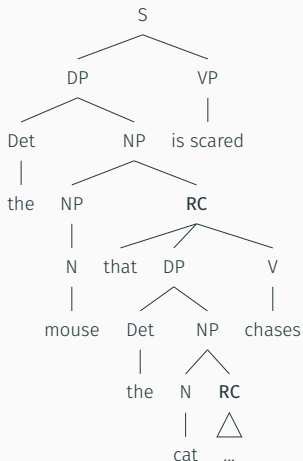
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Challenge: cross-serial dependencies

(55) Swiss German:

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...we Hans.DAT the house.ACC help paint.

‘... we helped Hans paint the house.’

- In Swiss German, predicates and their argument can be all **jumbled up!** Nevertheless, predicates overtly case-mark their arguments, with Dative (in the case of *help*) and Accusative (in the case of *paint*).
- If we wanted to design a context-free grammar keeping track of predicate-argument dependencies (and of the resulting case-marking patterns), we'd struggle in the case of Swiss German, precisely **because the dependencies cross**.
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https://colab.research.google.com/drive/11-deqVXa70AelTRruq7ZH3VXGflzkK_?usp=sharing